Middle East Technical University
Department of Electrical and Electronics Engineering

EE201 – Circuit Theory I

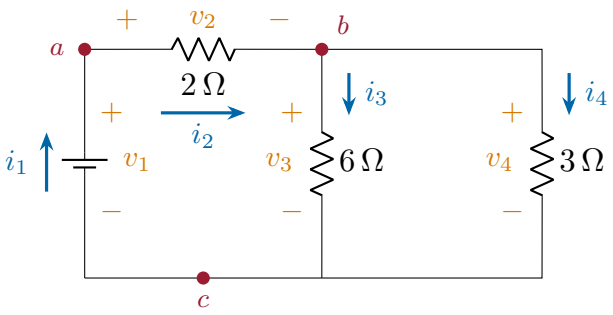
Lecture 2: Kirchhoff's Laws

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Colour code used in these notes and in the slides			
charge q	current i	voltage v	power p, energy w

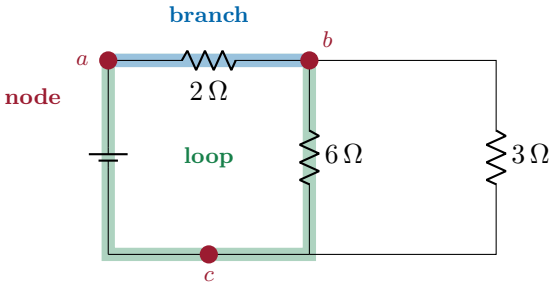
1. A circuit, and the question it asks

Here is a circuit. Every element in it is one you already know, and each reference direction has been chosen once and for all.



1.1. Node, branch, loop

Three words, all from Lecture 1, and every one of them is about the *drawing*, not about the components.



Node	$n = 3$	a connection point. Everything joined by ideal wire is *one* node, however far the drawing spreads it out. Here: a, b, c.
Branch	$b = 4$	one element, joining two nodes. Here: the source and the three resistors.
Loop	—	a closed path that passes through no node twice. Here: the left-hand one, the right-hand one, and the outer one.

We will use n for the number of nodes and b for the number of branches throughout.

1.2. What we do not know

Each branch carries a voltage and a current, and none of the eight is known yet:

$$\underbrace{v_1,\ v_2,\ v_3,\ v_4}_{\text{4 unknowns}} \qquad \underbrace{i_1,\ i_2,\ i_3,\ i_4}_{\text{4 unknowns}} \quad \Longrightarrow \quad \boxed{2b = 8 \text{ unknown functions of } t}$$

1.3. Half the equations are easy

Each element brings one equation of its own, a **terminal equation**. It describes that element *in isolation* and knows nothing about the rest of the circuit:

$$v_1 = 12\,\mathrm{V}, \qquad v_2 = 2\,i_2, \qquad v_3 = 6\,i_3, \qquad v_4 = 3\,i_4.$$

Element	Symbol	Terminal equation
resistor	R	$v = R\,i$
inductor	L	$v = L\,di/dt$
capacitor	C	$i = C\,dv/dt$
ind. voltage source	v_s	$v = v_s(t)$, i free
ind. current source	i_s	$i = i_s(t)$, v free

That is $b = 4$ equations for $2b = 8$ unknowns. **Exactly half.** And no amount of staring at the elements will produce the other half: put the same four components together differently and every current changes, while not one terminal equation does.

1.4. The missing half comes from the wiring

Terminal equations
one per element
b equations

depend on what *each element is, not on how it is wired*

+

Network equations
KCL and KVL
b equations

depend on how *things are wired, not on what the elements are*

The whole strategy of circuit theory

Cut the problem in two. The **elements** know nothing about the circuit they sit in; the **interconnection** knows nothing about the elements it joins. Kirchhoff's two laws are that second half, and this lecture is about them.

Because they ignore the elements, KCL and KVL hold for *every* lumped circuit — linear or nonlinear, time-invariant or time-varying, at DC or at 100 MHz.

2. Kirchhoff's Current Law

KCL

For **every node** of a lumped circuit and at **every instant** t, the algebraic sum of the currents *leaving* the node is zero:

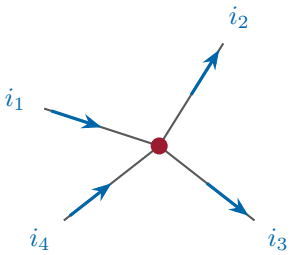
$$\sum_{k\,\in\,\text{node}} \pm i_k(t) \;=\; 0, \qquad \text{sign } + \text{ if the arrow points } \textit{away} \text{ from the node.}$$

Equivalently: *what flows in, flows out.*

Why. A node is a junction of ideal wires; it has no volume and stores no charge. If $q_{\text{node}}(t) = 0$ for all t, then charge conservation gives

$$\frac{dq_{\text{node}}}{dt} \;=\; \sum_{\text{in}} i_k - \sum_{\text{out}} i_k \;=\; 0.$$

So KCL is conservation of charge plus the lumped assumption from Lecture 1 (*nothing piles up*).



Leaving positive:

$$-i_1 + i_2 + i_3 - i_4 = 0$$

Entering positive gives the same equation times -1. Either convention works — what matters is using *one* of them all the way round.

Example 2.1 — KCL at a node

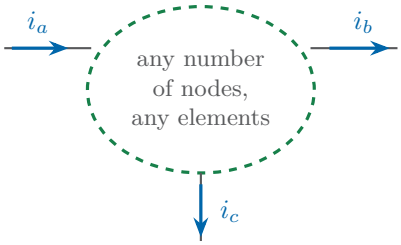
Four branches meet at a node with the reference arrows above. Measurements give $i_1 = 3\,\text{A}$, $i_2 = -2\,\text{A}$ and $i_3 = 6\,\text{A}$. Find i_4.

$$-i_1 + i_2 + i_3 - i_4 = 0 \implies -3 + (-2) + 6 - i_4 = 0 \implies \boxed{i_4 = 1\,\text{A}}$$

Two lessons. The measured $i_2 = -2\,\text{A}$ enters the equation *with its own sign*; we do not redraw the arrow. And the equation is written from the picture, not from any idea of which way the charge "really" goes.

2.1. KCL on a closed surface

Nothing in the argument above used the fact that the region enclosed was a single point. Draw *any* closed surface: if the charge inside it does not accumulate, the algebraic sum of the currents crossing it is zero.

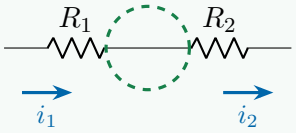


$$-i_a + i_b + i_c = 0$$

Whatever is inside — one node or a thousand, resistors or a whole amplifier — only the branches that *cross* the surface appear.

Example 2.2 — two consequences of the surface form

(a) Series elements carry the same current. Wrap a surface around the junction of two elements joined end to end. Exactly two branches cross it, so $i_1 = i_2$: a chain of elements has *one* current, and that is the definition of "in series".



$-i_1 + i_2 = 0$

(b) A two-terminal box returns everything it is given. Put the surface around an entire sub-network reached by two wires: the current in at one terminal equals the current out at the other, no matter how complicated the inside is. This is what lets us speak of "*the* current through" a black box.

Two things KCL is not

- It is an equation about **one instant**. $\sum i_k(t) = 0$ holds for each t separately; nothing is averaged over time.
- It says nothing about voltages, resistances or power. Only topology and current.

3. Kirchhoff's Voltage Law

KVL

For **every loop** of a lumped circuit and at **every instant** t, the algebraic sum of the branch voltages around the loop is zero:

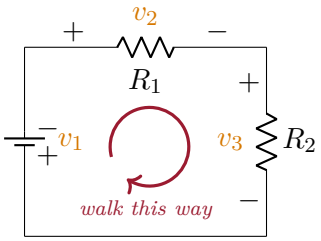
$$\sum_{k \in \text{loop}} \pm v_k(t) \;=\; 0, \qquad \text{sign } + \text{ if you traverse the branch from } + \text{ to } -.$$

Equivalently: *go all the way round and you come back to where you started.*

Why. In a lumped circuit every node j can be given a single number e_j, its potential with respect to a chosen reference. A branch voltage is then a difference, $v_k = e_{(+)} - e_{(-)}$, and going round a loop $a \to b \to c \to a$ the sum telescopes:

$$(e_a - e_b) + (e_b - e_c) + (e_c - e_a) \;=\; 0.$$

Energetically: carry a charge once around the loop and it ends where it began, so the net energy exchanged per unit charge is zero. Like KCL, this is the lumped model at work — it needs assumption (b), no changing magnetic field through the loop.[1]



$$-v_1 + v_2 + v_3 = 0$$

The source is traversed − to +, so it enters with a minus sign; the two resistors are traversed + to − and enter with a plus. Walking the other way multiplies everything by −1.

Voltage does not depend on the path

KVL is the same statement as: *the voltage between two nodes is the same whichever way you go between them.* If path 1 gives v_{ab} and path 2 gives v'_{ab}, then path 1 followed by path 2 backwards is a loop, so $v_{ab} - v'_{ab} = 0$. That is why "the voltage across R_2" is a meaningful phrase and why node potentials exist at all.

Example 2.3 — one loop

$v_s = 12\,\text{V}$, $R_1 = 2\,\Omega$, $R_2 = 4\,\Omega$ in the figure above. With the loop current i taken clockwise, the terminal equations give $v_2 = R_1 i$ and $v_3 = R_2 i$, and $v_1 = v_s$. KVL:

$$-12 + 2i + 4i = 0 \;\Longrightarrow\; \boxed{i = 2\,\text{A}}, \qquad v_2 = 4\,\text{V}, \quad v_3 = 8\,\text{V}.$$

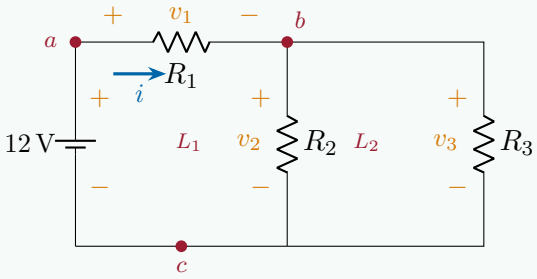
Check: $4 + 8 = 12\,\text{V}$. The $12\,\text{V}$ supplied is shared between the two resistors, the larger one taking the larger share — a fact we will soon turn into a formula.

[1] A transformer breaks that assumption on purpose: the loop picks up a voltage belonging to no element in the drawing. The cure is to draw the coupling as an element of its own, after which KVL holds again exactly as written.

3.1. A loop that is not a mesh, and what "independent" means

Example 2.4 — three loops, two equations

Take the circuit of page 1 with $v_s = 12\,\text{V}$, $R_1 = 2\,\Omega$, $R_2 = 6\,\Omega$, $R_3 = 3\,\Omega$.



Three loops can be drawn: L_1, L_2, and the outer one $L_3 = L_1 \cup L_2$. Walking each clockwise,

$$L_1:\ -12 + v_1 + v_2 = 0, \qquad L_2:\ -v_2 + v_3 = 0, \qquad L_3:\ -12 + v_1 + v_3 = 0.$$

The third is the sum of the first two: it is *true*, but it is not *new*. Adding it to the system would gain nothing.

Note also what L_2 says on its own: $v_2 = v_3$. Two elements joined at *both* ends always carry the same voltage — there is a loop containing nothing else, and KVL closes it. (Section 5 solves this circuit completely.)

4. How many equations, and which ones?

Recall from Section 1.1: n is the number of **nodes**, b the number of **branches**. For the circuit of page 1, $n = 3$ and $b = 4$.

Source of equations	**How many**	**Why that many**
KCL, one per node	$n - 1$	the n-th is the sum of the others
KVL, one per loop	$b - n + 1$	$= \ell$, the independent loops
network equations	**b**	
terminal equations, one per element	**b**	
total	**2b**	= number of unknowns v_k, i_k

Why is one KCL redundant? Add the node equations of *every* node: each branch leaves one node and enters another, so its current appears once with + and once with − and cancels. The sum is $0 = 0$, so the n equations carry only $n - 1$ pieces of information.

For the running example: $n = 3$, $b = 4$, so 2 independent KCL equations, 2 independent KVL equations, 4 terminal equations — 8 equations for the 8 unknowns $v_1..v_4$, $i_1..i_4$. We will soon turn this bookkeeping into systematic recipes (node and mesh analysis) that never write all $2b$ equations at once.

Some diagrams have no solution at all

KCL and KVL do not only solve circuits; they also decide which drawings *mean* anything.

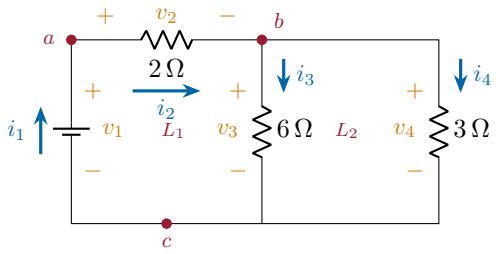
- Connect an ideal 5 V source directly across an ideal 9 V source. That loop contains nothing else, so KVL reads $5 - 9 = 0$. False — and no choice of currents can repair it.
- Put an ideal 2 A source in a chain with an ideal 3 A source. KCL at the node between them reads $2 = 3$. False again, whatever the voltages do.

Neither is an algebra mistake. The equations are contradictory, so the diagram describes no circuit at all. What has gone wrong is the *model*: an ideal source promises to hold its value no matter what current or voltage that costs, and two such promises can simply disagree.

Real components never promise that much. Give each battery its internal resistance and the equations become consistent again — and then tell you that a very large current circulates. Sometimes that is a harmless detail; sometimes it is the reason one of the batteries makes smoke.

5. Solving the circuit of page 1

Collect everything. Four terminal equations, two KCL, two KVL — eight equations, eight unknowns, and no method beyond substitution.



4 terminal equations	**2 KCL** $(n-1)$	**2 KVL** $(b-n+1)$
$v_1 = 12$	node a: $-i_1 + i_2 = 0$	L_1: $-v_1 + v_2 + v_3 = 0$
$v_2 = 2\,i_2$	node b: $-i_2 + i_3 + i_4 = 0$	L_2: $-v_3 + v_4 = 0$
$v_3 = 6\,i_3$		
$v_4 = 3\,i_4$		

Eliminate in the order the equations suggest. L_2 gives $6i_3 = 3i_4$, so $i_4 = 2i_3$. KCL at b then gives $i_2 = i_3 + i_4 = 3i_3$. Substituting into L_1,

$$-12 + 2i_2 + 6i_3 = 0 \quad \text{with} \quad i_3 = \tfrac{1}{3}i_2 \qquad \Longrightarrow \qquad -12 + 2i_2 + 2i_2 = 0.$$

Hence

$$i_1 = i_2 = 3\,\text{A}, \quad i_3 = 1\,\text{A}, \quad i_4 = 2\,\text{A}, \qquad v_1 = 12\,\text{V}, \quad v_2 = 6\,\text{V}, \quad v_3 = v_4 = 6\,\text{V}.$$

A free check: power must balance

The source delivers $12 \times 3 = 36\,\text{W}$; the resistors absorb $2(3)^2 + 6(1)^2 + 3(2)^2 = 18 + 6 + 12 = 36\,\text{W}$. So $\sum_k p_k = 0$, as Lecture 1 promised. One line, and it catches most sign errors.

The eight-equation system always works — it is essentially what a circuit simulator solves — but it is slow by hand, and it grows fast: a circuit with thirty branches would need sixty equations. We will soon learn much better ways of getting the same answer, which write only a handful of equations and still satisfy KCL and KVL automatically.

Summary

Law	Statement	Rests on
KCL	$\sum \pm i_k = 0$ at every node, every instant	conservation of charge; nothing piles up
KVL	$\sum \pm v_k = 0$ round every loop, every instant	node potentials exist; no stray flux

- A circuit gives $2b$ unknowns and $2b$ equations: b **terminal** equations (what the elements are) and b **network** equations, $n-1$ from KCL and $b-n+1$ from KVL (how they are wired).
- KCL holds on **any closed surface**, not only at a node. Elements joined end to end carry one current; a two-terminal box returns what it takes.
- KVL is the statement that **voltage does not depend on the path**. Elements joined at both ends carry one voltage.
- Signs come from the *reference* arrows and polarities you drew, never from a guess about which way things really go.
- Neither law knows anything about the elements: both hold for nonlinear and time-varying circuits exactly as written.

Next lecture

The other half of the equations: what a two-terminal element can be. Independent voltage and current sources, the general definition of a resistor, and the vocabulary for describing one — linear, bilateral, passive, controlled, time-invariant.